DS Lab 3: Descriptive Statistics

Measures of Central Tendency and Variability

In [10]:
```
import pandas as pd
import numpy as np
from sklearn import datasets
```

In [11]:
```
# Load Iris dataset
iris = datasets.load_iris()
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df['species'] = iris.target
df['species'] = df['species'].map({0:'Iris-setosa',1:'Iris-versicolor',2:'Iris-v
df.head()
```

Out[11]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

In [12]:
```
# Group by species and calculate statistics
grouped = df.groupby('species')

summary = grouped.agg(['mean','median','min','max','std'])
summary
```

Out[12]:

	sepal length (cm)					sepal width (cm)					
	mean	median	min	max	std	mean	median	min	max	std	n
species											
Iris-setosa	5.006	5.0	4.3	5.8	0.352490	3.428	3.4	2.3	4.4	0.379064	1
Iris-versicolor	5.936	5.9	4.9	7.0	0.516171	2.770	2.8	2.0	3.4	0.313798	4
Iris-virginica	6.588	6.5	4.9	7.9	0.635880	2.974	3.0	2.2	3.8	0.322497	5

In [13]:
```
# Separate statistics for each species
setosa = df[df['species']=='Iris-setosa']
versicolor = df[df['species']=='Iris-versicolor']
virginica = df[df['species']=='Iris-virginica']

print("Setosa Mean:\n", setosa.mean(numeric_only=True))
print("\nVersicolor Mean:\n", versicolor.mean(numeric_only=True))
print("\nVirginica Mean:\n", virginica.mean(numeric_only=True))
```

```
Setosa Mean:
 sepal length (cm)    5.006
sepal width (cm)     3.428
petal length (cm)    1.462
petal width (cm)     0.246
dtype: float64

Versicolor Mean:
 sepal length (cm)    5.936
sepal width (cm)     2.770
petal length (cm)    4.260
petal width (cm)     1.326
dtype: float64

Virginica Mean:
 sepal length (cm)    6.588
sepal width (cm)     2.974
petal length (cm)    5.552
petal width (cm)     2.026
dtype: float64
```

In [14]:
```
# Percentiles
df.quantile([0.25, 0.5, 0.75], numeric_only=True)
```

Out[14]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)
0.25	5.1	2.8	1.60	0.3
0.50	5.8	3.0	4.35	1.3
0.75	6.4	3.3	5.10	1.8

In [15]:
```
# Mode
df.mode()
```

Out[15]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species
0	5.0	3.0	1.4	0.2	Iris-setosa
1	NaN	NaN	1.5	NaN	Iris-versicolor
2	NaN	NaN	NaN	NaN	Iris-virginica

In [16]:
```
# Range = max - min
range_values = df.max(numeric_only=True) - df.min(numeric_only=True)
range_values
```

Out[16]:
```
sepal length (cm)    3.6
sepal width (cm)     2.4
petal length (cm)    5.9
petal width (cm)     2.4
dtype: float64
```

In [17]:
```
# Convert categorical to numerical (One-Hot Encoding)
encoded = pd.get_dummies(df['species'])
encoded.head()
```

Out[17]:

	Iris-setosa	Iris-versicolor	Iris-virginica
0	True	False	False
1	True	False	False
2	True	False	False
3	True	False	False
4	True	False	False

In [18]:
```
# Combine encoded data
df_encoded = pd.concat([df, encoded], axis=1)
df_encoded.head()
```

Out[18]:

	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	species	Iris-setosa	Iris-versicolor	Iris-virginica
0	5.1	3.5	1.4	0.2	Iris-setosa	True	False	False
1	4.9	3.0	1.4	0.2	Iris-setosa	True	False	False
2	4.7	3.2	1.3	0.2	Iris-setosa	True	False	False
3	4.6	3.1	1.5	0.2	Iris-setosa	True	False	False
4	5.0	3.6	1.4	0.2	Iris-setosa	True	False	False